PAPER ID—21131

B. Tech. (Computer Science and Engineering) (Artificial Intelligence and Machine Learning)

EXAMINATION, 2024

(Fourth Semester)

COMPUTER NETWORKS

Time : 3 *Hours* *Maximum Marks :* 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. **1** is compulsory. All questions carry equal marks.

1. (a) Differentiate between half-duplex and full-duplex transmission.

(b) Differentiate between packet and circuit switching.

(c) What is framing ?

(d) What is subnetting ?

(e) Differentiate between switch and router.

(f) What is flooding ?

(g) What is peer-to-peer communication ? 2×7=14

Unit I

2. Explain the following : 14

(a) Distributed processing

(b) Connection-less and connection-oriented services.

3. Explain the TCP/IP model in detail. 14

Unit II

4. What is the MAC address ? Explain the sliding window protocol in detail. 14

5. Explain and compare IPV4 and IPV6. 14

Unit III

6. How do routing algorithms help in network communication ? Also, explain Distance Vector Routing. 14

7. What is Process-to-Process communication ? Explain TCP connection management. 14

Unit IV

8. What are Quality of Service (QoS) parameters in computer networks ? Explain Qo Simproving techniques. 14

9. Explain the following : **14**

(a) e-MAIL

(b) Symmetric and asymmetric key cipher.